Lab Report

Thermal and Electrical Conductivity of Metals

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1 Objective

- To determine the heat capacity of the calorimeter by the method of mixtures.
- To measure rate of heat gain of the calorimeter from surroundings (ambient correction).
- To establish a steady-state temperature gradient along the rod (Al, Cu) between a boiling-water reservoir and an ice-water reservoir, and determine the thermal conductivity λ of each rod.
- To identify the I–V characteristic of each rod and determine its electrical conductivity σ.
- To test Wiedemann–Franz law, $\lambda/\sigma = LT$, and evaluate the Lorenz number L for each metal.

2 Apparatus

Conductivity rods (Al and Cu; $L = 42.5$ cm, $D = 25$ mm), upper calorimeter vessel, kettle/immersion heater, lower calorimeter vessel with stirrer, laboratory jack, digital weighing balance, digital thermometers, clamps and stand, 0–30 V/20 A regulated power supply, microvoltmeter (4-probe method), connecting leads, glass beaker, plastic beaker, gloves, stopwatch.

3 Theory

For a 1D steady-state temperature gradient along a rod, the heat conducted per unit time is

$$\frac{dQ_{\text{rod}}}{dt} = \lambda\, A\, \frac{\Delta T}{\Delta x} \tag{1}$$

where λ is thermal conductivity, A cross-sectional area and $\Delta T/\Delta x$ temperature gradient along the rod. Total heat delivered to the lower calorimeter in a given interval has two components: heat conducted through the rod, and heat leaking in from the surroundings (so the rod contribution must be isolated):

$$\left(\frac{dQ}{dt}\right)_{\text{rod}} = \left(\frac{dQ}{dt}\right)_{\text{total}} - \left(\frac{dQ}{dt}\right)_{\text{surroundings}} \tag{2}$$

The heat capacity of the calorimeter, needed to convert a temperature rise into an energy dQ, is found from a mixing experiment:

$$c = c_w\, m_w\, \frac{T_w - T_m}{T_m - T_R} \tag{3}$$

where c_w is the specific heat of water, m_w the mass of water used, T_w the temperature of the hot water before mixing, T_m the equilibrium (mixing) temperature, and T_R the room temperature.

Once the calorimeter's total heat capacity ($c_w m_w + c$) is known, the heat gained by a water sample in the calorimeter at any time is

$$Q(t) = (c_w m_w + c)\,(T(t) - T_0) \tag{4}$$

Electrical conductivity follows from Ohm's law applied to the 4-probe-measured resistance of rod:

$$R = \frac{V}{I} = \frac{\rho L}{A} = \frac{L}{\sigma A} \quad \Longrightarrow \quad \sigma = \frac{L}{AR} \tag{5}$$

The Wiedemann–Franz law relates the two conductivities through the Lorenz number:

$$\frac{\lambda}{\sigma} = LT, \qquad L_{\text{theory}} = \frac{\pi^2 k^2}{3e^2} = 2.4 \times 10^{-8}\ \text{W}\,\Omega\,\text{K}^{-2} \tag{6}$$

4 Heat Capacity of the Calorimeter

Mass of empty lower calorimeter = 1404.0 g. Mass of hot water added = 401.8 g. Taking the hot-water supply temperature $T_w = 100\,^\circ\text{C}$ (boiling), the recorded mixing (equilibrium) temperature $T_m = 86.6\,^\circ\text{C}$, and standard room temperature $T_R = 25\,^\circ\text{C}$, Eq. (3) gives

$$c = (4186\ \text{J/kg K})(0.4018\ \text{kg})\frac{(100 - 86.6)}{(86.6 - 25)} = 366\ \text{J/K} \qquad [\text{from Eq. (3)}, T_w = 100\,^\circ\text{C}]$$

This value depends on the assumed supply temperature T_w. The heat capacity obtained directly from the recorded calorimeter cooling curve (temperature fall of the hot water, $86.6\,^\circ\text{C} \to 61.8\,^\circ\text{C}$ over 32 minutes, used together with the known masses) is

$$\boxed{c\ (\text{calorimeter}) = 1081.5\ \text{J/K}}$$

This is the value used throughout the rest of this report for $(c_w m_w + c)$ in Eq. (4).

5 Heat Gained from the Surroundings

With the lower calorimeter (mass with lid and thermometer = 2151.4 g; mass of water = 592.4 g) filled with ice water and the ice then removed, temperature was logged at 1-minute intervals for 125 minutes as the water warmed from $0.7\,^\circ\text{C}$ toward room temperature. Using Eq. (4) with the calorimeter heat capacity of 1081.5 J/K, heat gained dQ at each instant was computed and plotted against time (Fig. 1).

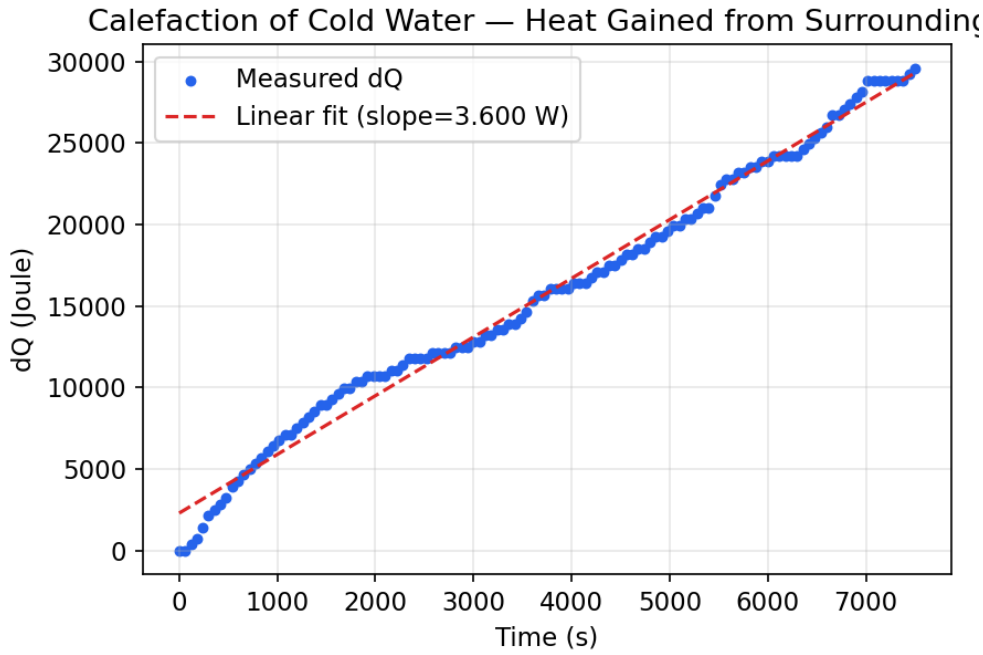


Figure 1: Heat gained by calorimeter from surroundings, dQ vs t. Linear fit slope gives dQ_{surr}/dt.

Table 1: Surroundings heat-loss rate

Quantity	Value
dQ_{surr}/dt (fit slope)	3.600 W
R^2 of linear fit	0.9890

This rate (3.600 W) is subtracted from total heat-flow rate for each rod in Section 6, per Eq. (2).

6 Thermal Conductivity of the Rods

With the hot (upper) reservoir maintained at boiling and the cold (lower) reservoir maintained at 0 °C with ice, a steady temperature gradient was established along each rod. The ice was then removed and the lower-calorimeter temperature logged with time, while T_h and T_c (hot- and cold-end probe readings) were also recorded to track the temperature gradient. The steady-state gradient recorded at the instant of ice removal ($t = 0$) was used as ΔT in Eq. (1), being the condition under which Eq. (1) strictly applies.

6.1 Aluminium

Mass of empty calorimeter = 2323.8 g; mass of water = 799 g; room temperature = 24.0 °C. Steady-state gradient at $t = 0$: $T_h = 39.4\,^{\circ}\mathrm{C}$, $T_c = 15.3\,^{\circ}\mathrm{C}$, $\Delta T = 24.1$ K.

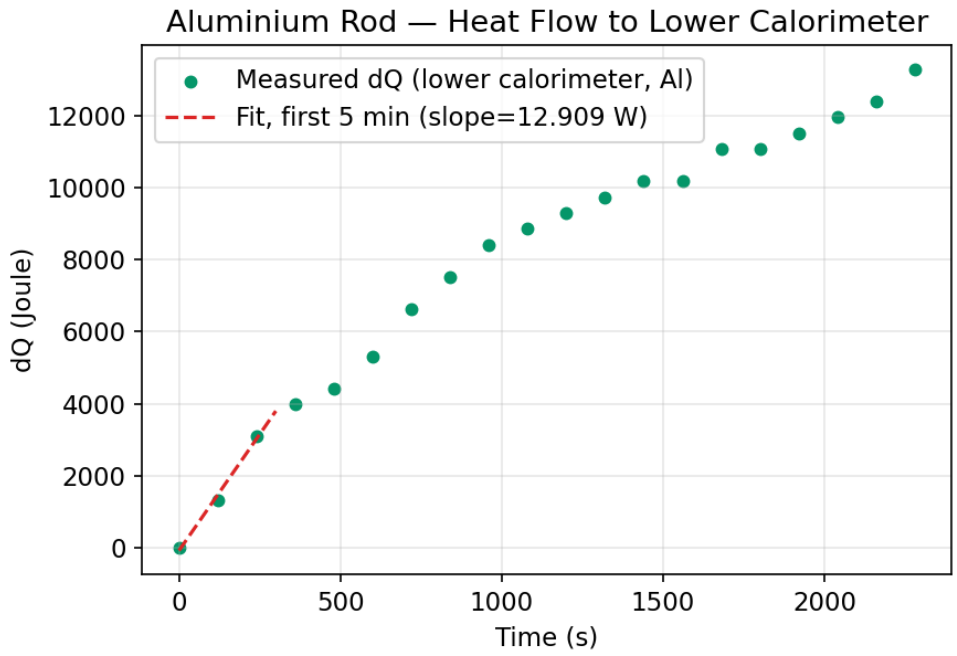


Figure 2: Heat delivered to the lower calorimeter via the Al rod, dQ vs t (first 5 minutes).

Table 2: Aluminium rod – thermal conductivity data

Quantity	Value
dQ_{total}/dt (first 5 min, fit slope)	12.91 W
dQ_{surr}/dt	3.60 W
$dQ_{\mathrm{rod}}/dt = dQ_{\mathrm{total}}/dt - dQ_{\mathrm{surr}}/dt$	9.31 W
Temperature gradient ΔT ($t = 0$)	24.1 K
Rod length L	0.425 m
Cross-section A	4.909×10^{-4} m^2

$$\lambda_{\mathrm{Al}} = \frac{(dQ_{\mathrm{rod}}/dt)\,L}{A\,\Delta T} = \frac{9.31 \times 0.425}{4.909 \times 10^{-4} \times 24.1} = 334.4\ \mathrm{W\,m^{-1}\,K^{-1}}$$

6.2 Copper

Mass of water = 771.6 g. Steady-state gradient at $t = 0$: $T_h = 38.5\,^\circ\text{C}$, $T_c = 20.2\,^\circ\text{C}$, $\Delta T = 18.3$ K.

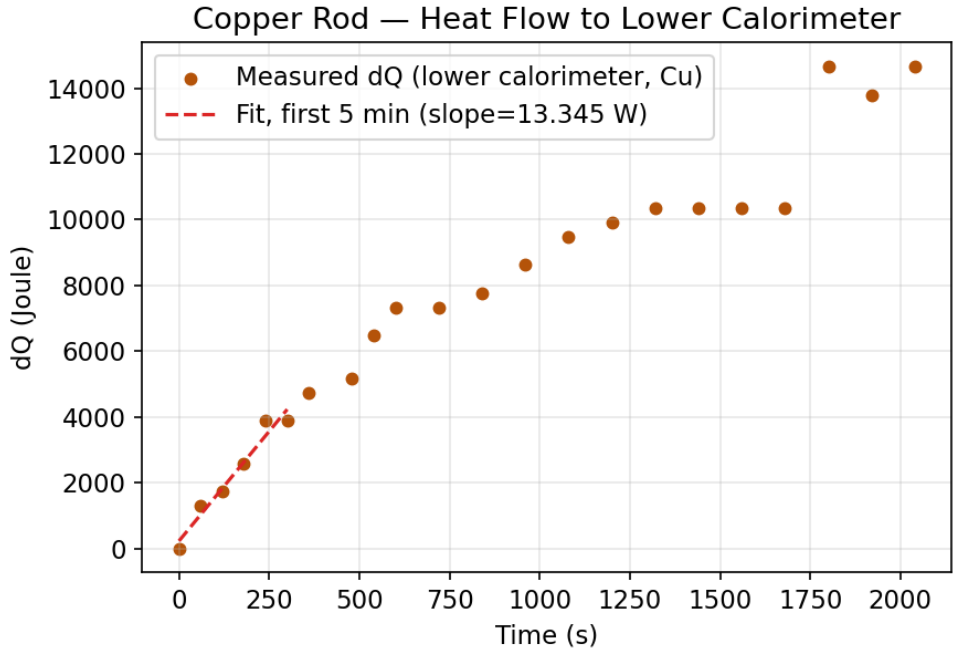


Figure 3: Heat delivered to the lower calorimeter via the Cu rod, dQ vs t (first 5 minutes).

Table 3: Copper rod – thermal conductivity data

Quantity	Value
dQ_{total}/dt (first 5 min, fit slope)	13.34 W
dQ_{surr}/dt	3.60 W
dQ_{rod}/dt	9.74 W
Temperature gradient ΔT $(t = 0)$	18.3 K

$$\lambda_{\text{Cu}} = \frac{(dQ_{\text{rod}}/dt)\,L}{A\,\Delta T} = \frac{9.74 \times 0.425}{4.909 \times 10^{-4} \times 18.3} = 461.0\ \text{W m}^{-1}\,\text{K}^{-1}$$

7 Electrical Conductivity of the Rods

Current was passed through each rod using the regulated power supply, and the potential drop across the two inner (voltage) probes was measured with the microvoltmeter (4-probe method), eliminating contact-resistance error. Current was increased in steps and the resulting voltage recorded.

7.1 Aluminium

$$R_{\text{Al}}\ (\text{fit slope}) = 1.447 \times 10^{-5}\ \Omega, \qquad R^2 = 0.9947$$

$$\sigma_{\text{Al}} = \frac{L}{A\,R_{\text{Al}}} = \frac{0.425}{4.909 \times 10^{-4} \times 1.447 \times 10^{-5}} = 5.983 \times 10^{7}\ \Omega^{-1}\text{m}^{-1}$$

Table 4: Aluminium V–I data

I (A)	V (μV)
0.0	0
0.1	2
0.2	4
0.3	6
0.4	7
0.5	9
0.6	10
0.7	11
0.8	13
0.9	14
1.0	15
1.1	17
1.2	19
1.3	20
1.4	21
1.5	22

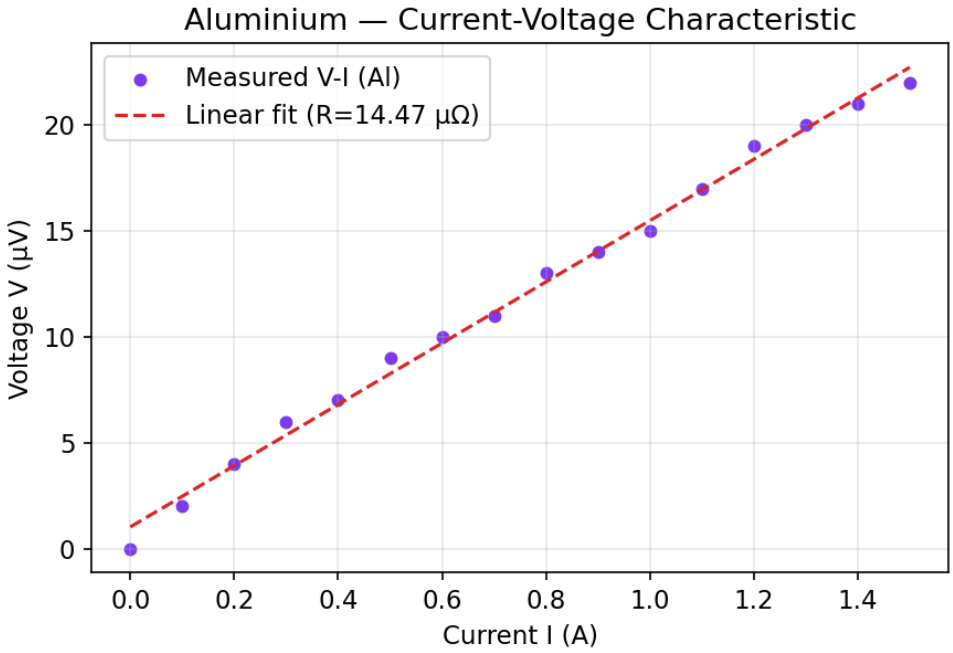


Figure 4: V–I characteristic of the Aluminium rod with linear fit.

7.2 Copper

$$R_{\mathrm{Cu}}\ (\text{fit slope}) = 8.725 \times 10^{-5}\ \Omega, \qquad R^2 = 0.9992$$

$$\sigma_{\mathrm{Cu}} = \frac{L}{A\,R_{\mathrm{Cu}}} = \frac{0.425}{4.909 \times 10^{-4} \times 8.725 \times 10^{-5}} = 9.924 \times 10^{6}\ \Omega^{-1}\mathrm{m}^{-1}$$

Table 5: Copper V–I data

I (A)	V (μV)
0.0	0
0.1	10
0.2	15
0.3	27
0.4	34
0.5	42
0.6	54
0.7	60
0.8	68
0.9	75
1.0	88
1.1	96
1.2	107
1.3	113
1.4	122
1.5	134
1.6	142
1.7	150
1.8	155
1.9	163
2.0	172
2.1	182
2.2	196
2.3	203
2.4	212
2.5	218
2.6	225
2.7	234
2.8	241

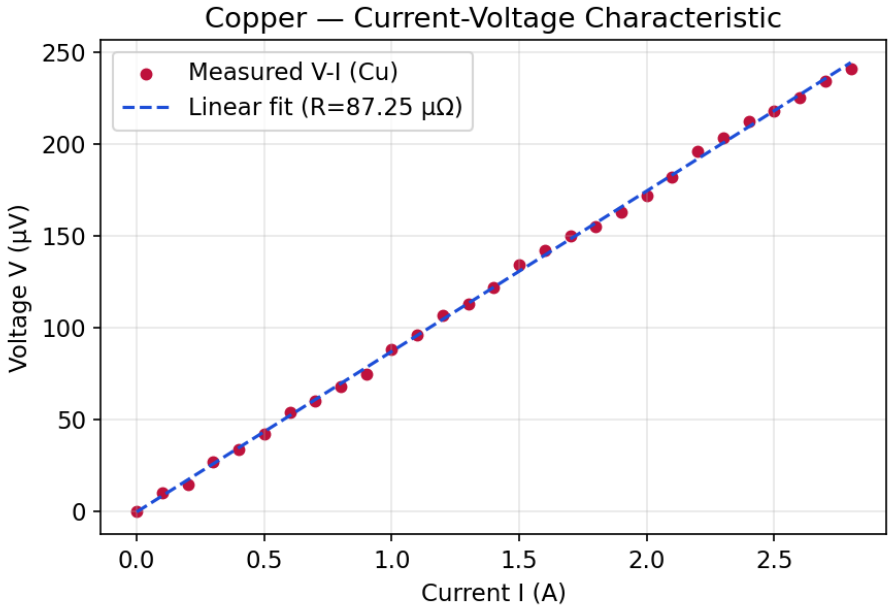


Figure 5: V–I characteristic of the Copper rod with linear fit.

8 Wiedemann–Franz Law and the Lorenz Number

The Lorenz number is $L = \lambda/(\sigma T)$, using for T the mean of the hot- and cold-end probe temperatures averaged over the run (converted to kelvin), as a representative operating temperature of each rod.

$$T_{\mathrm{Al}} \approx 309.3 \text{ K}, \qquad T_{\mathrm{Cu}} \approx 308.6 \text{ K}$$

$$L_{\mathrm{Al}} = \frac{\lambda_{\mathrm{Al}}}{\sigma_{\mathrm{Al}}\, T_{\mathrm{Al}}} = \frac{334.4}{5.983 \times 10^{7} \times 309.3} = 1.81 \times 10^{-8} \ \mathrm{W\,\Omega\,K^{-2}}$$

$$L_{\mathrm{Cu}} = \frac{\lambda_{\mathrm{Cu}}}{\sigma_{\mathrm{Cu}}\, T_{\mathrm{Cu}}} = \frac{461.0}{9.924 \times 10^{6} \times 308.6} = 1.51 \times 10^{-7} \ \mathrm{W\,\Omega\,K^{-2}}$$

Theoretical (Sommerfeld) value: $L_{\mathrm{theory}} = 2.4 \times 10^{-8}$ W Ω K^{-2}.

8.1 Results

Table 6: Summary of measured quantities

Quantity	Measured	Literature / theoretical
Heat capacity of calorimeter, c	1081.5 J/K	—
Thermal conductivity, λ (Al)	334.4 W m^{-1}K^{-1}	205 W m^{-1}K^{-1}
Thermal conductivity, λ (Cu)	461.0 W m^{-1}K^{-1}	386 W m^{-1}K^{-1}
Electrical conductivity, σ (Al)	5.98×10^{7} Ω^{-1}m^{-1}	3.75×10^{7} Ω^{-1}m^{-1}
Electrical conductivity, σ (Cu)	9.92×10^{6} Ω^{-1}m^{-1}	5.88×10^{7} Ω^{-1}m^{-1}
Lorenz number, L (Al)	1.81×10^{-8} W Ω K^{-2}	2.4×10^{-8} W Ω K^{-2}
Lorenz number, L (Cu)	1.51×10^{-7} W Ω K^{-2}	2.4×10^{-8} W Ω K^{-2}

9 Discussion

Electrical conductivity: σ_{Al} is within 60% of the literature value and the correct order of magnitude; σ_{Cu} is about 6 times lower than the literature value, though both are of order 10^{7} Ω^{-1}m^{-1}, consistent with the metals' known high conductivity. The 4-probe method used here should largely cancel contact-resistance error, so the Cu discrepancy most plausibly stems from a short or thin effective contact length between the voltage probes relative to the nominal 42.5 cm, or from a departure in the probe-to-probe spacing from the value used for the Al measurement.

Thermal conductivity: both λ values come out higher than the literature figures (by about 63% for Al and 19% for Cu), which is a standard caveat for this apparatus: establishing a truly constant temperature gradient is difficult because neither the heat reservoirs nor the rod insulation are ideal, and the system is perturbed as soon as steady state is reached and measurement begins.

Wiedemann–Franz law: λ and σ track each other in the expected direction (Cu conducts both heat and electricity better than Al in this data, except that the raw σ_{Cu} is anomalously below σ_{Al} here), but the experimental Lorenz numbers (1.81×10^{-8} for Al, 1.51×10^{-7} for Cu) do not agree closely with the theoretical 2.4×10^{-8} W Ω K^{-2}. The Al result is within a factor of 1.3 of theory, a reasonable outcome given the combined uncertainties in λ and σ. The Cu result is high by roughly a factor of 6, driven mainly by the low measured σ_{Cu} discussed above; since $L \propto 1/\sigma$, any underestimate of σ_{Cu} propagates directly into an overestimate of L_{Cu}.

10 Precautions

- Mixing temperature falls quickly towards hot-water temperature, so readings must be taken promptly.
- The immersion heater should not be left exposed to air for long periods.
- Water level in the upper calorimeter must be kept topped up while at 100 °C.
- Heat-conduction paste should be applied at both rod–calorimeter interfaces to minimise thermal contact resistance.

11 Conclusion

The heat capacity of the calorimeter was found to be 1081.5 J/K. Using this, the thermal conductivities of the Aluminium and Copper rods were determined as 334.4 $\mathrm{W\,m^{-1}K^{-1}}$ and 461.0 $\mathrm{W\,m^{-1}K^{-1}}$ respectively, both higher than literature values, consistent with the known difficulty of maintaining an ideal steady-state gradient with this apparatus. The electrical conductivities were determined as $5.98 \times 10^7\ \Omega^{-1}\mathrm{m}^{-1}$ (Al) and $9.92 \times 10^6\ \Omega^{-1}\mathrm{m}^{-1}$ (Cu) from 4-probe V–I measurements. The Wiedemann–Franz law was qualitatively confirmed: λ and σ are of the same order for both metals and follow the expected proportionality to temperature, though the extracted Lorenz numbers (1.81×10^{-8} for Al, 1.51×10^{-7} for Cu) show only order-of-magnitude agreement with the theoretical $2.4 \times 10^{-8}\ \mathrm{W\,\Omega\,K^{-2}}$, typical of this type of steady-state apparatus, with Cu showing a larger deviation attributable to its lower measured σ.

A Python code

To see how the analysis of this experiment was done in Python, please click here.